

percent, 18 percent to 23 percent, and 28 percent to 33 percent all have the highest frequency of 7.

Consider the annual returns in Exhibit 8 for three portfolios for Portfolios P, Q and R. Portfolios P and R were created in Year 1, Portfolio Q was created in Year 2.

Exhibit 8: Annual Portfolio Returns

	Year 1 (%)	Year 2 (%)	Year 3 (%)	Year 4 (%)	Year 5 (%)
Portfolio P	-3.0	4.0	5.0	3.0	7.0
Portfolio Q	NA	-3.0	6.0	4.0	8.0
Portfolio R	1.0	-1.0	4.0	4.0	3.0

3. The median annual return for:

- A. Portfolio P is 4.5 percent.
- B. Portfolio Q is 4.0 percent.
- C. Portfolio R is higher than its arithmetic mean annual return.

Solution:

C is correct. The median of Portfolio R is 0.8 percent higher than the mean for Portfolio R. A is incorrect because the median annual return for Portfolio P is 4.0 percent. B is incorrect because the median annual return for Portfolio Q is 5.0 percent (midpoint of 4 percent and 6 percent).

4. The mode for Portfolio R is:

- A. 1.0 percent.
- B. 3.0 percent.
- C. 4.0 percent.

Solution:

C is correct. The mode is the most frequently occurring value in a dataset, which for Portfolio R is 4.0 percent.

A fund had the following returns over the past 10 years:

Exhibit 9: Fund Returns for 10 Years

Year	Return
1	4.5%
2	6.0%
3	1.5%
4	-2.0%
5	0.0%
6	4.5%
7	3.5%
8	2.5%